Math Practice Test — Measuring detectives

Year 3 · AV3 · 3rd term (practice)

Name: _____ Date: _____ Class: _____ Score: _____

Practice test made from the topics taught in class (Toddle class diary). The AV3 study guide has not been published yet. It is not an official school test.

Read each question carefully. Remember: 1 L = 1,000 mL · 1 kg = 1,000 g · 1 m = 100 cm. In the Problems part, show your work.

Block 1 of 3 · about 10 minutes · then a break

Part A — Multiple choice

3 points

Circle the correct answer.

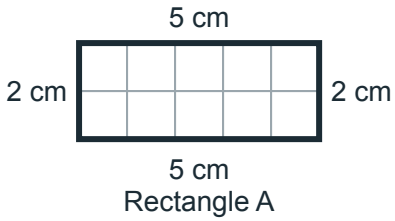
1. Which unit would you use to measure the length of the classroom?
   a. centimeters
   b. meters
   c. grams
   d. liters

2. Which unit would you use to measure the mass of a paper clip?
   a. grams
   b. kilograms
   c. liters
   d. meters

3. How many milliliters are in **2 liters**?
   a. 20 mL
   b. 200 mL
   c. 2,000 mL
   d. 1,000 mL

4. Which is the **heaviest**?
   a. 500 g
   b. 1 kg
   c. 900 g
   d. 100 g

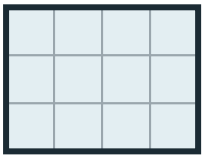
5. What is the **perimeter** of this rectangle?



Rectangle A

a. 7 cm
b. 10 cm
c. 14 cm
d. 12 cm

6. What is the **area** of this rectangle?



Rectangle B

a. 7 squares
b. 12 squares
c. 14 squares
d. 10 squares

Part B — Problems

0.6 points

Show your work. Then write the answer in a complete sentence.

7. Mom has a 1-liter bottle of water. She pours 300 mL into a glass. How many milliliters are left in the bottle?

Break: stand up, drink some water and come back.

Block 2 of 3 · about 10 minutes · then a break

Part C — True or false

1.5 points

Write T (true) or F (false). If it is false, write the correct sentence.

8. ( ) 1 kilogram is the same as 1,000 grams.
   Correct it: _____

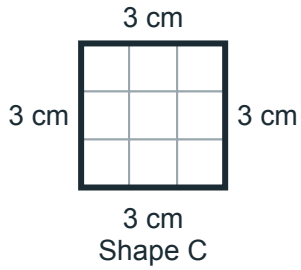
9. ( ) We measure milk in kilograms.
   Correct it: _____

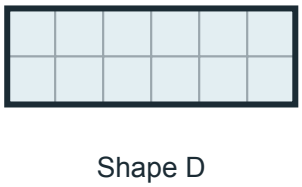
10. ( ) 1 meter is the same as 10 centimeters.
    Correct it: _____

11. ( ) The area is the space inside a shape.
    Correct it: _____

12. ( ) To measure with a ruler, we start at number 1.
    Correct it: _____

Look at the shapes





Shape C: a square with sides of 3 cm. Shape D: count the squares.

Part D — Perimeter and area

2 points

Write the answer with the unit.

13. What is the perimeter of Shape C?

    _____

14. What is the area of Shape C?

    _____

15. What is the area of Shape D?

    _____

16. Shape D: each square has sides of 1 cm. What is its perimeter?

    _____

17. How many centimeters are in 2 meters?

Break: stand up, drink some water and come back.

Block 3 of 3 · about 10 minutes · last block

Part E — Complete the sentences

1.5 points

Use the words in the word bank. One word is extra.

Word bank: liters · milliliters · kilograms · grams · centimeters · perimeter

18. A big bottle of juice holds 2 _____ .

19. A spoon of medicine holds 5 _____ .

20. A bag of rice has a mass of 5 _____ .

21. My pencil is 15 _____ long.

22. The distance **all the way around** a shape is the _____ .

Part F — What are we measuring?

0.8 points

Put an X in the column of the unit you would use.

	liters / mL	**kg / g**	**m / cm**
water in a fish tank			
a watermelon			
the height of a door			
a crayon			

Part G — Problems

0.6 points

Show your work. Then write the answer in a complete sentence.

23. A garden is a rectangle 6 m long and 3 m wide. Leo wants to put a fence **all the way around** it. How many meters of fence does he need?

Good luck! · Total: 10

Answer key (Gabarito) — Math Practice Test — Measuring detectives

> Para quem aplica: a prova vem em 3 blocos curtos, com pausa. Leia a instrução com ele antes de começar cada bloco; deixe-o se levantar e se mexer entre os blocos.

Block 1

Part A — Multiple choice

1. b) meters
2. a) grams
3. c) 2,000 mL — *1 L = 1,000 mL, so 2 L = 2,000 mL*
4. b) 1 kg — *1 kg = 1,000 g*
5. c) 14 cm — *5 + 2 + 5 + 2 = 14*
6. b) 12 squares — *3 rows of 4 = 12*

Part B — Problems

7. **Expected answer (example):** There are 700 mL left in the bottle (1 L = 1,000 mL; 1,000 – 300 = 700).
   Para o responsável, confira: troca 1 L por 1.000 mL; conta certa (700 mL); resposta em frase (0,6)

Block 2

Part C — True or false

8. T (true)
9. F (false) — We measure milk in liters (or milliliters).
10. F (false) — 1 meter is the same as 100 centimeters.
11. T (true)
12. F (false) — To measure with a ruler, we start at zero.

Part D — Perimeter and area

13. 12 cm — *3 + 3 + 3 + 3*
14. 9 squares — *3 rows of 3*
15. 12 squares — *2 rows of 6*
16. 16 cm — *6 + 2 + 6 + 2*
17. 200 cm — *1 m = 100 cm; 100 + 100*

Block 3

Part E — Complete the sentences

18. liters
19. milliliters
20. kilograms
21. centimeters
22. perimeter

Part F — What are we measuring?

water in a fish tank: liters / mL
a watermelon: kg / g
the height of a door: m / cm
a crayon: m / cm

Part G — Problems

23. **Expected answer (example):** He needs 18 meters of fence (6 + 3 + 6 + 3 = 18).
    Para o responsável, confira: entende que "all the way around" é o perímetro; soma os 4 lados (18 m); resposta em frase (0,6)